Energy yield loss during communication outage

Wind Farm T (demo) · 01.09.2026 – 03.09.2026 · 3 turbines (T1, T2, T3)

ESTIMATED LOST ENERGY (P50)

311.0 MWh

LIKELY RANGE (P10 – P90)

272.7 – 355.5 MWh

Total for the wind farm, 3 turbines over 72 h. Value: EUR 41,539 (Day-ahead market) · EUR 41,539 (EEG market premium) · EUR 22,745 (PPA (fixed price)).

1 Outage details

Wind farm	Wind Farm T (demo)
Operator / insured	Demo Operator GmbH
Insurer · policy	Demo Insurance AG · DEMO-0000-00
Outage start	01 September 2026 at 00:00 (local time, Europe/Berlin)
Outage end	04 September 2026 at 00:00 (local time, Europe/Berlin)
Duration	72 hours
Affected turbines	T1, T2, T3 (4.5 MW rated power each)
Suspected cause	A production stop is suspected; possible causes include cable damage between the wind farm and the grid connection point or a technical fault preventing feed-in. The root cause is under investigation.

P50 is the best estimate; the actual lost energy is expected to fall between P10 and P90 in 8 out of 10 cases, based on the method's historical errors. Totals are summed over all affected turbines (P10/P90 conservatively, since the turbines share the same reference data). Energy = expected power (kW, power curve) × 10 min.

2 Reconstructed wind and lost production

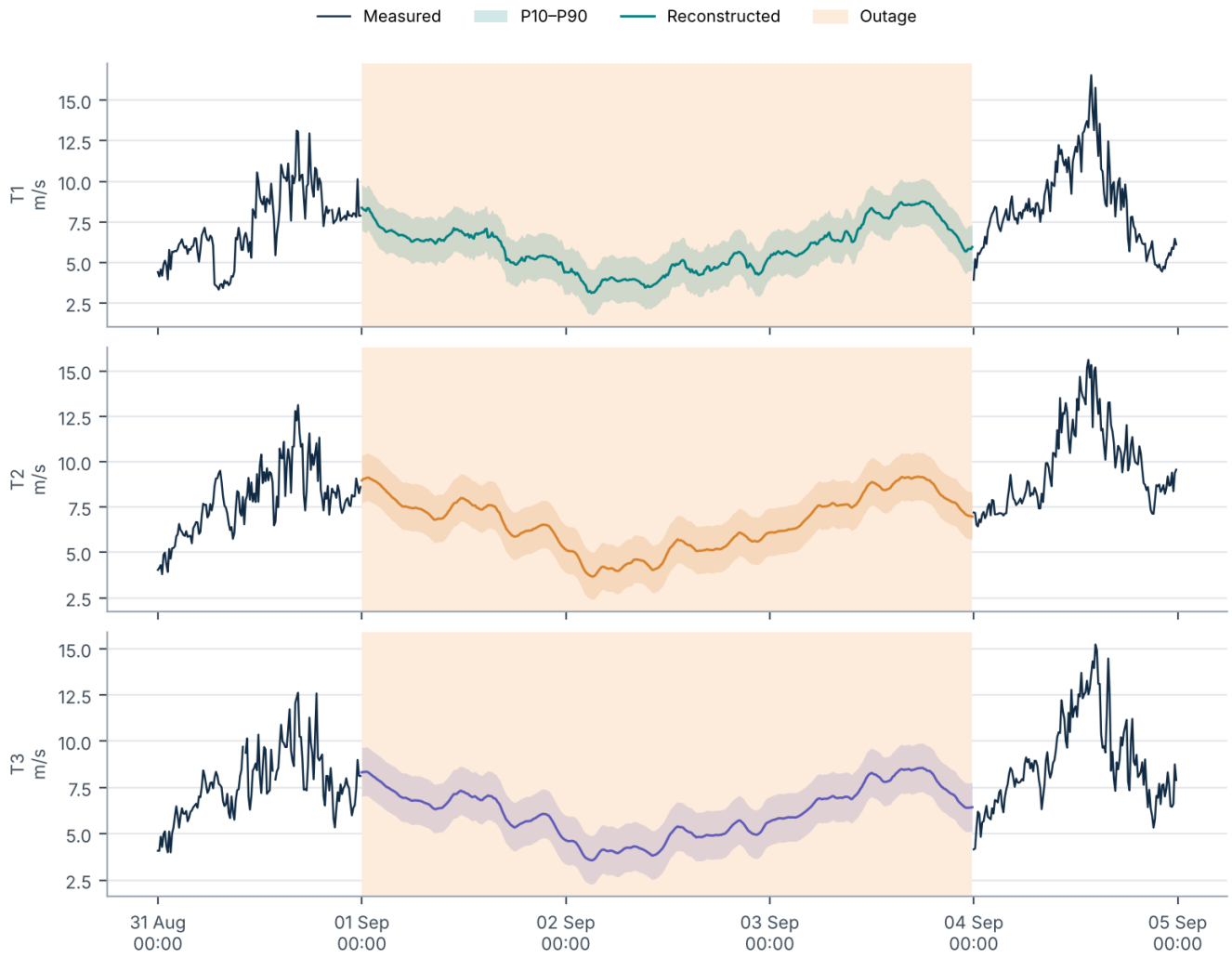


Figure 1 – Measured wind speed at each turbine (navy) one day before and after the outage, and the reconstructed wind speed during the outage (shaded) with its P10–P90 band.

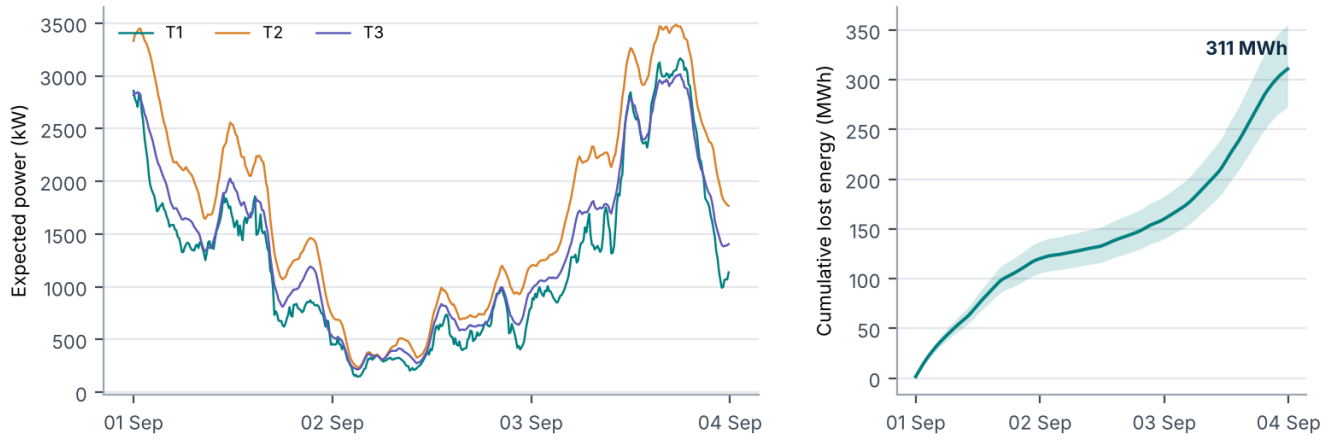


Figure 2 – Left: expected power per turbine during the outage. Right: cumulative lost energy for the wind farm with P10–P90 band.

3 Valuation

Which value applies depends on how the wind farm sells its electricity, so all three common remuneration models are shown.

Model	Price basis	EUR P50	EUR P10 – P90	EUR/MWh
Market	day-ahead price per interval	41,539	36,418 – 47,474	133.55
EEG premium	day-ahead price + 0.00 EUR/MWh premium	41,539	36,418 – 47,474	133.55
PPA	fixed 73.13 EUR/MWh	22,745	19,942 – 25,995	73.13

EUR/MWh = average value per lost MWh. EEG: under the market premium model the operator sells at the market and receives max(0, reference value − monthly market value of onshore wind) on top; no payment in negative-price intervals (§51 EEG). PPA: fixed pay-as-produced price.

Notes and assumptions

• EEG premium = max(0, reference value 73.30 EUR/MWh − monthly market value of onshore wind (2026-09: 109.40 EUR/MWh, German average, not the price of this outage)); the market value was above the reference value, so no premium was due and EEG equals the day-ahead value.
• Market benchmarks used where the contract price was not provided: EEG reference value 73.30 EUR/MWh, PPA price 73.13 EUR/MWh – to be replaced by the actual contract values.
• EEG reference value benchmark source: typical for plants now in operation: Bundesnetzagentur onshore wind tender Aug 2024, volume-weighted average award 7.33 ct/kWh.
• Source of the 2026-09 market value: official monthly market value (netztransparenz.de).
• PPA price benchmark source: German onshore wind pay-as-produced PPA benchmark, national average 73.13 EUR/MWh (Veyt, 2025).

Average day-ahead price in window	147.45 EUR/MWh
Negative-price intervals	0 (0.0 MWh)
Provisionally priced share	0%
Price data	historic price dataset in the repository; covers until 25 Oct 2026 23:45

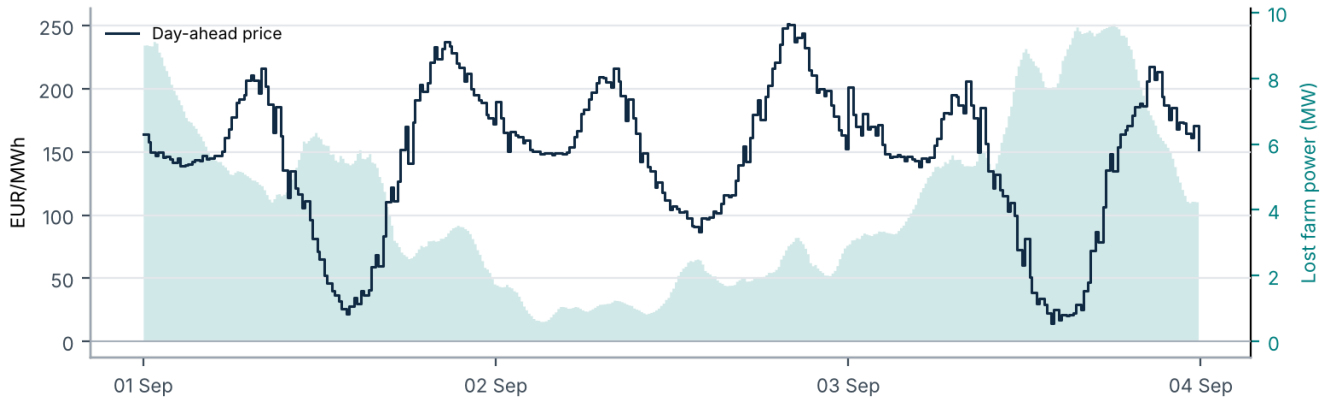


Figure 3 – Day-ahead price per interval (line) and lost farm power (shaded). Each 10-minute slice of lost energy is valued at the price of the interval it falls in. Day-ahead prices: Bundesnetzagentur | SMARD.de, via Energy-Charts (Fraunhofer ISE), licence CC BY 4.0.

4 Method, accuracy and limitations

For each affected turbine, the wind speed during the outage was reconstructed as a weighted sum of wind speeds measured at reference turbines in other wind farms (H1, H2, H3, H4, H5, O1, O2). Each reference is first shifted by the time delay that best matches the target turbine and smoothed over 130 minutes; all weights are forced to be non-negative. References and extra terms (wind-direction correction curve, wind direction, turbulence) were kept only where they improved accuracy in cross-validation over 3-day blocks of history. Lost energy is the turbine power curve averaged over the model's historical error distribution.

Why this method: every number can be checked by hand. Each reference has one weight and one time shift (Appendix A); non-negative weights mean stronger wind at a reference can never lower the estimate. Inputs are selected automatically, but only kept if they measurably improve accuracy on held-out history, so the model stays as simple as the data allows.

Turbine	Wind RMSE	Wind MAE	Power RMSE	|Energy err.| %	Val. wind RMSE	Val. energy %
T1	1.19	0.89	672	9.6	1.45	+16.4
T2	1.11	0.83	640	7.9	1.17	-0.7
T3	1.12	0.83	629	8.5	1.29	-2.2

First four columns: cross-validation over 29 historical 3-day blocks (model refitted without each block, then scored on it). 'Val.' columns: the 4–9 Sep validation week, held out entirely ('–' = not available for this data). Wind in m/s, power in kW, energy error = deviation of summed energy.

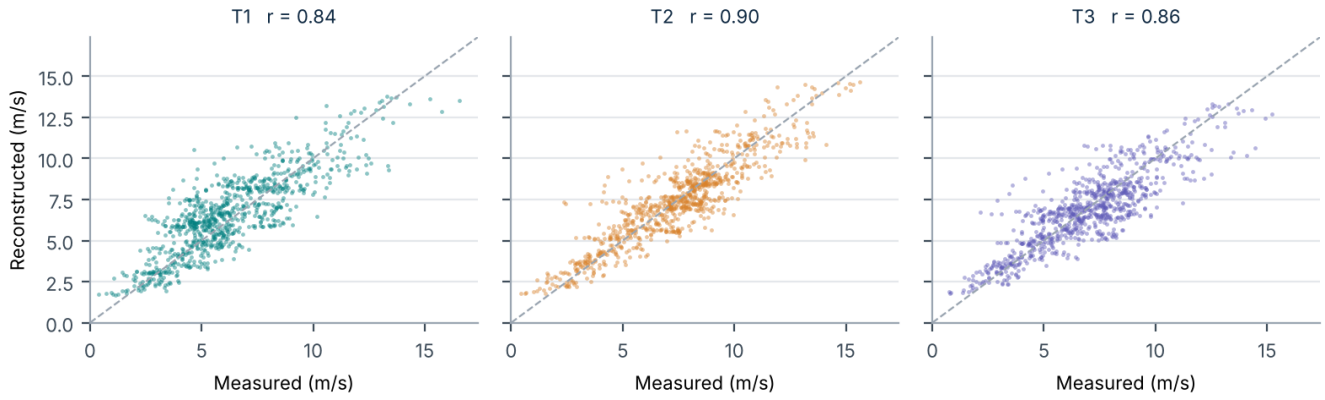


Figure 4 – Validation week: reconstructed vs measured 10-minute wind speed. Points on the dashed line would be perfect predictions.

Limitations

• The figure is an estimate. Individual 10-minute values are uncertain; totals over several days are considerably more reliable.
• It assumes the turbines were available and would have operated normally; curtailment, maintenance or grid limits during the outage are not included.
• Loss of communication does not by itself prove a production stop. The grid operator's metering data for the period should be used to confirm that no energy was fed in.
• Monetary value, waiting period and policy terms depend on contracts provided by the operator. Market valuation uses day-ahead prices; actual revenue may differ (balancing, marketing fees).

Appendix A Reproducibility and audit trail

Re-running the tool with the input files below (verified by their SHA-256 fingerprints) and the same model version reproduces every figure in this report.

Model	Bounded linear regression on lag-aligned reference turbines (git f0460fe)
Reference turbines	H1, H2, H3, H4, H5, O1, O2
Outage window	2026-09-01 00:00 – 2026-09-03 23:50 (Europe/Berlin, 10-min stamps)
Power curve	rated 4,500 kW from 11.5 m/s; first output at 3 m/s
Report generated	24 Oct 2026 14:39

Turbine	H1	H2	H3	H4	H5	O1	O2
T1	0.26 (-20)	0.18 (-20)	–	–	–	0.34 (+0)	–
T2	–	0.00 (-30)	0.15 (-30)	0.00 (-30)	0.22 (-30)	0.22 (+0)	0.17 (+10)
T3	–	0.13 (-30)	–	0.08 (-20)	0.18 (-30)	0.33 (+0)	–

Weight of each reference turbine's wind speed, with its time shift in minutes in brackets (negative = the wind reaches the target turbine first). Example: 0.40 means +1 m/s at that reference adds 0.40 m/s at the target turbine. '–' = reference not used for this turbine.

Turbine	Extra terms (coefficient)	Intercept m/s	CV R^2	Train rows
T1	dir sin +0.38, dir cos +0.08, turb. +0.49, dir. curve (6)	0.30	0.69	12212
T2	turb. +0.76	0.13	0.76	12411
T3	turb. +0.59	0.27	0.71	12411

Breakdown by turbine	P10 MWh	P50 MWh	P90 MWh
T1	77.6	89.8	101.9
T2	111.0	122.2	142.9
T3	84.1	99.1	110.7
Total	272.7	311.0	355.5

Input file	SHA-256 fingerprint
H1_data.xlsx	`f00af7198126ac3a5839ab4cb1a09eb6ec54bf2bc98f7485b7ccd1afbf1334c5`
H2_data.xlsx	`5dfa885703128edcdddf2b4c7729f22219fd5803648f27c6a73737024bea0365`
H3_data.xlsx	`ddb005bfaf46af048cd5e752c63c78fae566c20f786be152872ba5178b3e2c4b`
H4_data.xlsx	`f4a213f36e8400ec36d444d2e53ef01d67658384f2078a4c8c334c861364caf6`
H5_data.xlsx	`ffa19bb61b99c3ae5347dac1c671e20955c70f466d1851466edf7b58cfc1722e`
O1_data.xlsx	`7139e3380bb1f1acc4f9094a7e372f03f50d7fbd06178c2a4ac269a7e541bfd7`
O2_data.xlsx	`d12d19e7a0c496c213cb24ce0a88c9290b212404b70b21b437a021857d0ea292`
T1_data.xlsx	`56e4139af0334f3e28393ac5757b545323790fc89043b0d4cbda08d400731b55`
T1_power_curve.xlsx	`cd7fbbf142148d9fcd84e95cfb9e2404c2e9a9bd399d13cf3dc669a4e123a6c1`
T2_data.xlsx	`3a6bb6e248a8ce67fd224a85ac433b09b48d89479b6044a92f43569f9a76b9e3`
T2_power_curve.xlsx	`2cfcffb092bb7412deb12ee0332b4e7d8200ea9768f211cd0961c18ee03f389e`
T3_data.xlsx	`2aba8062d668474ace743c0fc4c79c802137af72d63f43edd0d91b3e30d5fc4a`
T3_power_curve.xlsx	`7434f64acf9b77bfefd084f4b26d25a0cd86a136206bf21048ef4548a50a6f02`